



SoundWave

Power BI Business Intelligence Project Master HSIM 2025-2026

**Sahar Haghshenas
Laura Hun**

Introduction

The SoundWave Power BI report transforms raw streaming and revenue data into a structured, decision-oriented analysis of platform performance. The objective is to provide clear insights into overall activity, artist contribution, and monetization efficiency.

The report is organized around five key business questions, each corresponding to a dedicated page: overall performance (Executive Dashboard), streaming evolution (Streaming Dynamics), artist contribution (Artist Performance), revenue generation (Monetization & Royalties), and growth opportunities (Strategic Insights).

Technically, the report is built using a star-schema data model in Power BI. Data was cleaned and standardized in Power Query, and DAX measures were created to calculate key indicators such as total streams, growth rates, and revenue metrics. Visuals were selected based on analytical relevance to ensure clarity and logical progression.

Overall, the dashboard converts raw data into actionable insights, supporting performance monitoring and strategic decision-making.

2. Data Preparation (Power Query)

2.1 Cleaning Process

Given the heterogeneity of the SoundWave dataset, a structured data preparation phase was conducted using Power Query before any modeling or visualization was implemented. The raw data was provided in multiple formats, including Excel (.xlsx), CSV (.csv), CSV with alternative delimiters (;), TSV (.txt), and PDF documentation. Because these formats differ in structure, encoding, and delimiter conventions, an initial harmonization step was required to ensure consistent integration into Power BI.

Excel files such as *artists.xlsx*, *tracks.xlsx*, and *subscribers.xlsx* were imported using the native Excel connector. Header rows were verified and promoted where necessary, and column types were inspected to avoid incorrect automatic detection. CSV files required more careful handling: some used standard comma delimiters, while others relied on semicolons or Latin-1 encoding. In these cases, the correct delimiter and encoding were explicitly specified during import to prevent column misalignment or character corruption. The TSV file (*royalty_payments.txt*) was imported by defining the tab delimiter manually. Although the PDF quarterly report was not integrated into the data model, it was used as contextual documentation to validate high-level financial trends.

Once all files were imported, the primary objective of the cleaning process was to ensure numerical consistency, correct data typing, and reliable time structure.

A major issue concerned financial fields, particularly *gross_amount_usd* and *net_amount_usd*. These columns contained formatting artifacts such as leading currency symbols (\$) and

thousand separators (commas), which caused Power BI to interpret them as text rather than numeric values. Through Power Query transformations, currency symbols and separators were removed using the Replace Values function.

The columns were then converted from Text to Decimal Number. This correction was essential to enable accurate aggregation, prevent calculation errors, and support DAX measures such as Total Revenue, Revenue per Stream, and growth rates. Without these adjustments, financial KPIs would either fail or produce unreliable results.

A second data quality issue involved the *monthly_listeners* field in the artists table. This column contained mixed formatting, including numeric values with thousand separators and potential non-numeric entries. To standardize the field, commas were removed and non-numeric values were replaced with nulls where appropriate. The column was subsequently converted to Whole Number. This ensured correct numerical sorting, reliable ranking, and accurate positioning in the Monthly Listeners vs Streams scatter analysis.

Time-related data required particular attention. The quarter field stored values such as “Q2 2024,” combining both period and year in a single text column. This format is unsuitable for chronological sorting and time-intelligence calculations. The column was therefore split into two components: Quarter_Label and Year. The year component was converted to numeric format. Rather than relying solely on this textual quarter information, a dedicated Date dimension table was created and linked to all fact tables via the date column. This allowed the implementation of a structured time hierarchy (Year → Quarter → Month → Day) and enabled quarter-over-quarter and month-over-month growth calculations using DAX time-intelligence functions.

Additional standardization steps were performed to ensure relational integrity. Country names were harmonized (e.g., converting abbreviations such as “US” to “United States”) and assigned the appropriate data category (Country/Region) to enable correct geographical mapping. Primary and foreign key columns such as artist_id, track_id, and album_id were verified to ensure consistent formatting across tables. Null values were reviewed and handled appropriately to avoid broken relationships in the data model.

Overall, the cleaning and transformation phase was critical to converting heterogeneous raw files into a structured analytical dataset. By correcting financial formatting, standardizing numeric fields, restructuring time attributes, and harmonizing keys across Excel, CSV, and TSV sources, the dataset was transformed into a reliable foundation for modeling and DAX calculations. This preparation phase reflects a fundamental BI principle: analytical reliability depends on data consistency before visualization.

3. Data Model

To support clear analysis, robust time intelligence, and strong performance in Power BI, a **star schema data model** was implemented. This structure places a single, central fact table at the core of the model, surrounded by several dimension tables that provide descriptive context such

as artists, tracks, albums, subscribers, and time. The star schema is particularly well suited for analytical workloads because it simplifies relationships, enables efficient aggregation, and scales cleanly as new KPIs are introduced.

Fact Table

daily_streams (Primary Fact)

The **daily_streams** table is the central fact table of the model. Each row represents a daily observation of platform activity at the chosen analytical grain (for example, per track, per artist, per country, or per platform, depending on available fields).

Typical fields include: date, track_id, artist_id, country, platform, streams_count, gross_amount_usd, net_amount_usd, quarter or related time attributes

This table drives the core performance analysis of the report, including: Total Streams, Revenue KPIs, Time-based trends, Top artists, Country comparisons, Growth calculations. It acts as the quantitative backbone of the entire dashboard.

Dimension Tables

Artists: The artists table contains descriptive attributes for each artist, including: artist_id, artist_name, monthly_listeners, country_origin, label_id, etc. It is used to segment and filter streaming data by artist characteristics and powers visuals such as: Top Artists rankings, Monthly Listeners vs Streams scatter plot, Label-based performance analysis.

Tracks : The **tracks** table provides metadata for individual tracks: track_id, track_name, artist_id, album_id, genre, duration_sec, release_date, etc. It enables genre-level analysis and supports visuals such as: Streams by Genre, Performance by catalog segment. Content-based segmentation

Albums : The albums table contains album-level information: album_id, album_name, release_year, label_id, genre. It enriches track data and allows aggregation at album level or analysis by catalog age.

Subscribers: The subscribers table provides contextual information about the platform's user base: date, total_subscribers, active_subscribers, plan_type, etc. Although not central to every page, it allows performance normalization, such as: Streams per subscriber, Revenue per active subscriber, Growth relative to user base evolution

Date_Table – Time Normalization Layer

A dedicated **Date_Table** was created using the DAX **CALENDAR (or CALENDARAUTO)** function. This table includes a continuous sequence of dates covering the full period of analysis and contains at minimum: Date, Year, Quarter, Month, Year-Month, Week, Day of week.

The Date_Table was explicitly marked as the official Date Table in Power BI.

Beyond supporting time-intelligence functions, the Date_Table serves an essential **normalization function** in the model. Instead of relying on fragmented or text-based time attributes (such as “Q2 2024” stored in fact tables), all temporal analysis is centralized and standardized within this dedicated dimension.

This normalization ensures: Consistent time hierarchies (Year → Quarter → Month → Day)

, Proper chronological sorting, Compatibility with DAX time-intelligence functions such as:

DATEADD, TOTALYTD, Quarter-over-quarter growth

By centralizing time logic in a single dimension, the model avoids inconsistencies and guarantees analytical coherence across all report pages.

Key Relationships

In this star schema, one-to-many relationships run from each dimension to the central fact table:

- Date_Table[Date] → daily_streams[date]
- artists[artist_id] → daily_streams[artist_id]
- tracks[track_id] → daily_streams[track_id]
- albums[album_id] → tracks[album_id] (then indirectly to daily_streams through tracks)
- subscribers[date] → Date_Table[Date] (for subscriber-based metrics)

Filters flow from the dimension tables (one side) down to the fact table (many side). This directional structure ensures intuitive filtering behavior and prevents ambiguous filter paths.

Model Justification

The choice of a star schema and the use of daily_streams as the central fact table are justified by several principles.

First, the model ensures **clear one-to-many relationships**. Each dimension connects to the fact table through a single key, reducing complexity and minimizing the risk of modeling errors.

Second, it enables **efficient filtering and aggregation**. By separating descriptive attributes from quantitative measures, Power BI's VertiPaq engine can compress and aggregate the model efficiently.

Third, it avoids **circular relationships and ambiguity**. The star schema prevents complex filter propagation paths that could otherwise lead to incorrect DAX results.

Finally, the central role of `daily_streams` is conceptually coherent. Since most business questions in the report revolve around how much content is streamed, by whom, and when, it is logical to position this table as the analytical hub. All major pages—Executive Dashboard, Streaming Dynamics, Artist Performance, Monetization, and Strategic Insights—derive their primary measures from this fact table.

Overall, the data model provides a normalized analytical foundation for the SoundWave Power BI report. By combining a central fact table, well-structured dimensions, and a dedicated time normalization layer, the model ensures reliable KPIs, consistent filtering, and scalable performance analysis.

4. DAX Measures and KPI

To transform the cleaned and structured data model into meaningful business intelligence, a set of core DAX measures was developed in Power BI. These measures constitute the analytical foundation of the report and are reused consistently across all dashboard pages, including the Executive Dashboard, Streaming Dynamics, Artist Performance, Monetization & Royalties, and Strategic Insights.

4.1 Volume and Growth Measures

Total Streams

The **Total Streams** measure aggregates streaming volume from the `daily_streams` fact table within the current filter context (date, country, genre, artist, etc.). This measure represents the primary indicator of platform engagement and serves as the quantitative backbone of the report. It is displayed prominently on the Executive Dashboard and feeds most trend, ranking, and segmentation visuals. From a business perspective, Total Streams reflects audience demand and platform activity intensity.

Streams Growth %

To move beyond static volume analysis, a growth indicator was implemented using DAX time intelligence. The logic compares streaming volume between the current period and the previous equivalent period (e.g., month-over-month or quarter-over-quarter).

4.2 Revenue and Monetization Measures

Total Gross Revenue

The **Total Gross Revenue** measure aggregates all gross revenue values from the relevant financial fields. It represents top-line monetization before deductions such as royalties or distribution fees. This KPI provides a high-level view of platform value generation.

Net Revenue

The **Net Revenue** measure sums retained revenue after payouts and operational costs. From a strategic standpoint, Net Revenue is more representative of actual platform profitability and operational efficiency. The comparison between Gross and Net Revenue enables margin analysis and assessment of payout structures.

Revenue per Stream

One of the most important indicators in the report is **Revenue per Stream**, defined as:

Net Revenue divided by Total Streams. In DAX, the logic is implemented using: Revenue per Stream =

`DIVIDE([Net Revenue], [Total Streams])`

This measure evaluates monetization efficiency independently of volume. While Total Streams captures demand, Revenue per Stream captures value extraction. It allows meaningful comparison across segments with different scales (e.g., high-volume vs low-volume countries) and helps identify structural changes in pricing, contracts, or monetization strategies.

4.3 Artist-Level Performance Measures

Streams per Artist

To normalize performance at the artist level, a measure was created dividing total streaming volume by the distinct number of artists within the current context. Conceptually:

$\text{Total Streams} \div \text{Distinct Count of Artists}$

This metric provides an average performance benchmark and helps determine whether a segment's success is concentrated among a few superstars or distributed across a broader base of artists. It supports deeper interpretation on the Artist Performance page and complements ranking visuals.

4.4 Time Intelligence and Period Logic

Time intelligence plays a central role in transforming static aggregates into dynamic insights. Using the dedicated Date_Table and functions such as: `DATEADD`, `SAMEPERIODLASTYEAR`, `TOTALYTD`. the model enables: Month-over-month growth analysis ,Quarter-over-quarter

comparisons, Year-over-year evaluation, Cumulative year-to-date tracking. These calculations allow the dashboard to highlight seasonal patterns, structural shifts, and emerging trends rather than merely reporting absolute values.

By shifting the filter context within the normalized time dimension, DAX ensures consistent and reliable period comparisons across all report pages.

Together, the DAX measures form a coherent performance framework built around five core dimensions: Volume (Total Streams), Growth (Streams Growth %), Value (Gross and Net Revenue), Efficiency (Revenue per Stream), Distribution (Streams per Artist)

The combination of structured modeling, reusable measures and transforms raw streaming records into a comprehensive analytical system capable of supporting both executive decision-making and detailed operational evaluation.

5. Visual Design Logic and Page Structure

5.1 Executive Dashboard

The Executive Dashboard is designed as the entry point of the report and provides a concise overview of SoundWave's overall performance. Its objective is to answer the managerial question: **"How is the platform performing?"** The layout follows a clear hierarchy, moving from global indicators to performance drivers.

At the top of the page, three KPI cards present the most important metrics: **Total Streams (10B)**, **Gross Revenue (40.3M)**, and **Revenue per Stream (\$0.00403)**. These indicators summarize platform activity (volume), financial performance (value), and monetization efficiency. Placing them at the top ensures immediate visibility and supports quick executive decision-making.

The **line chart** displays total streams by month. This visual was chosen because it is the most appropriate format for time-series analysis. It highlights seasonal variations and growth trends, allowing users to identify performance peaks and slowdowns throughout the year.

The **map visualization** shows streams by country using proportional bubbles. This format allows an intuitive understanding of geographic distribution and market concentration. It helps identify dominant regions and potential growth markets.

The **genre distribution chart** presents the share of streams across the top five genres. This visual explains content consumption patterns and shows whether performance depends on a single genre or is diversified.

Finally, the **bar chart of the Top 10 Artists by revenue** highlights performance concentration among key contributors. Ranking visuals were selected because they clearly display comparison and relative importance.

The Executive Dashboard therefore functions as a strategic overview before moving to more detailed analytical pages.

5.2 Streaming Dynamics

The *Streaming Dynamics* page analyzes how streaming activity evolves over time and across key segments. While the Executive Dashboard provides a global overview, this page focuses on understanding **how performance is generated and distributed**.

At the top of the page, two KPI cards are displayed: **Total Streams (6.1B)** and **Artist Growth (18%)**. These indicators summarize overall engagement within the selected context and highlight performance momentum. Including a growth metric allows the analysis to move beyond volume and assess acceleration or slowdown.

The **line chart (Total Streams by Month)** presents the evolution of streaming activity over time. A line chart was chosen because it is the most appropriate visual for time-series analysis. It clearly highlights seasonal variations and identifies peak periods (notably mid-year), suggesting possible release cycles or increased user engagement during certain months.

The **Top 5 Genre pie chart** illustrates the distribution of streams across main music categories. This visual was selected to show proportional contribution rather than precise comparison. It helps determine whether growth is concentrated in specific genres or diversified across content types, providing insight into audience preferences.

The **Total Streams by Country bar chart** compares streaming volumes across markets. A bar chart enables straightforward comparison between countries and reveals market balance or concentration. This helps identify leading markets and potential areas for expansion.

Finally, the **Daily Top Tracks table** introduces a dynamic perspective by displaying current rankings, previous positions, and peak rank. This table adds competitive context and highlights short-term performance shifts, complementing the aggregated metrics shown in other visuals.

This progression ensures that streaming behavior is analyzed from multiple complementary angles time, content, geography, and competition—providing a comprehensive understanding of demand dynamics.

5.3 Artist Performance

The *Artist Performance* page focuses on analyzing individual artist contribution to overall platform success. While previous pages examine global streaming dynamics, this section shifts the perspective to answer the question: **“Which artists drive performance?”**

A dynamic artist selector allows users to focus on a specific artist (e.g., *Silver Lining*). This interactive element was included to enable personalized analysis and comparison between artists.

At the top left, key performance indicators summarize the selected artist's profile: **Total Streams (43.5M)**, **Monthly Listeners (11M)**, **Social Followers (2.1M)**, and **Growth Trend (+13%)**. These KPIs combine audience size, engagement, and momentum, providing a complete overview of popularity and recent performance evolution.

The **line chart (Total Streams of the Artist)** displays monthly streaming trends. A time-series visual was chosen to identify performance fluctuations, release impacts, or seasonal peaks. It allows evaluation of whether the artist's growth is stable or volatile.

The **pie chart (Total Streams per Track)** shows the distribution of streams across main tracks. This visual highlights performance concentration and identifies the artist's most successful songs. It helps determine whether success is driven by one hit or by multiple strong tracks.

Similarly, the **donut chart (Total Streams by Album)** aggregates performance at album level. This provides insight into catalog strength and long-term value generation.

On the right side, the **Top 10 Artist ranking panel** allows quick comparison between leading performers. Ranking visuals are effective for showing relative positioning and competitive context within the platform.

5.4 Monetization & Royalties

The *Monetization & Royalties* page analyzes how streaming activity is converted into financial value. While previous sections focus on engagement and artist performance, this page answers the strategic question: **"How does streaming translate into revenue and payouts?"**

At the top of the page, key financial KPIs summarize overall monetization performance:

- **Total Net Royalties (31.2M)**
- **Total Gross Royalties (40.3M)**
- **Revenue per Stream (\$0.00312)**
- **Payments Paid (86%)**
- **Outstanding Amount (4.3M)**

These indicators provide a clear distinction between generated revenue (gross), retained value (net), monetization efficiency, and payout status. Positioning them at the top ensures immediate visibility of financial health.

The **Net Revenue by Label & Artist bar chart** compares earnings across labels. A horizontal bar chart was selected because it allows easy comparison between partners and highlights

revenue concentration. This visual helps identify dominant labels and assess dependency on major partners.

The **Payment Status pie chart** presents the proportion of paid, pending, and processing royalties. This visual provides operational insight into payout efficiency and transparency. It quickly communicates whether most obligations have been fulfilled.

The **Artist to Pay table** lists specific outstanding payments by artist. Including a table here is justified because it provides precise, actionable information that cannot be summarized visually. It supports operational follow-up and financial control.

Finally, the **Gross vs Net Revenue line chart** shows the evolution of generated versus retained revenue over time. A time-series comparison was chosen to illustrate margin dynamics and detect fluctuations between gross earnings and actual retained income.

5.5 Strategic Insights

The *Strategic Insights* page synthesizes the analysis presented in previous sections and shifts the focus from descriptive performance to strategic interpretation. Its objective is to answer the forward-looking question: **“Where are the main growth drivers and strategic opportunities?”**

At the top of the page, two KPI indicators highlight structural concentration:

- **Top 3 Artists = 5% of Total Revenue**
- **Top 1 Label = 9% of Net Revenue**

These metrics were included to measure dependency risk and revenue concentration. They provide an immediate understanding of how diversified or concentrated the platform's revenue structure is.

The **Artist Revenue Share treemap** visualizes revenue distribution across artists. A treemap was selected because it effectively represents proportional contribution within a large group. It quickly reveals dominant artists and the relative weight of smaller contributors, helping assess diversification and market balance.

The **Strategic Artist Positioning Matrix (scatter plot)** compares artists based on two key dimensions:

- **Total Streams (volume)**
- **Revenue per Stream (monetization efficiency)**

This visual was chosen because it enables multidimensional analysis. By combining volume and value, it distinguishes different strategic profiles:

- High streams / High revenue per stream → Strong strategic assets
- High streams / Low revenue per stream → High volume but low efficiency
- Low streams / High revenue per stream → Niche but highly profitable
- Low streams / Low revenue per stream → Development opportunities

This final page transforms analytical findings into managerial insight. It moves beyond reporting past performance and provides a framework for identifying growth priorities, partnership risks, and investment opportunities, while maintaining visual clarity and consistency. For example, if an artist has very high streams but low revenue per stream, it means they are popular but not very profitable. In that case, we might invest in better monetization strategies, like premium placements or new partnerships.

If an artist has low streams but high revenue per stream, it means they are small but very profitable. That could be a niche opportunity, so we might invest in marketing to increase their visibility.

We also included filters on every page. This allows users to interact with the dashboard and focus on specific countries, artists, genres, or time periods.

6. Limitations and Challenges

Although the SoundWave report provides structured and coherent insights, several limitations must be acknowledged.

First, the dataset is limited to a relatively short time period. This restricts the ability to identify long-term trends, multi-year seasonality, or structural changes in user behavior. The analysis therefore reflects short-term performance rather than long-term evolution.

Second, the dataset does not include detailed information about pricing strategies, subscription types, or regional pricing differences. As a result, indicators such as *Revenue per Stream* are interpreted under simplified assumptions and may not fully capture real-world monetization complexity.

Third, the absence of marketing, promotional, or user acquisition data prevents causal analysis. While the report identifies growth patterns and performance shifts, it cannot determine the exact drivers behind these changes.

Additionally, the data required significant cleaning due to inconsistent formats (currency symbols, mixed data types, time fields). Although these issues were corrected, data quality constraints may still affect precision.

Finally, the analysis remains descriptive and diagnostic. No predictive modeling or forecasting techniques were implemented, meaning the report supports understanding of past and current performance but does not estimate future outcomes.

Despite these limitations, the dashboard provides a reliable analytical framework for evaluating streaming activity, artist contribution, and monetization efficiency.

7.Conclusion

This Power BI analysis provides a structured and strategic overview of SoundWave's performance, transforming raw streaming and revenue data into a coherent decision-support system. By organizing the report into five complementary sections—Executive Dashboard, Streaming Dynamics, Artist Performance, Monetization & Royalties, and Strategic Insights—the analysis balances high-level visibility with detailed operational insight.

The findings indicate sustained streaming growth, a significant concentration of performance among top artists, relatively stable monetization efficiency per stream, and a revenue structure that remains dependent on key labels. These results highlight both the strengths of SoundWave's current market positioning and potential areas for diversification, risk mitigation, and strategic renegotiation.

Beyond descriptive reporting, the dashboard equips management with a dynamic monitoring tool. It enables identification of high-growth segments across artists, genres, and countries; evaluation of monetization efficiency; and prioritization of strategic initiatives in content investment, partnership development, and geographic expansion.

Future enhancements could extend the analytical depth of the project by integrating predictive modeling (e.g., stream and revenue forecasting), advanced user segmentation, and churn analysis. Such extensions would shift the report from primarily diagnostic analytics toward a more predictive and prescriptive decision framework.

Overall, the SoundWave Power BI report demonstrates how structured data modeling can convert raw operational data into actionable strategic intelligence.